

Case Name
Patents Kitty Porn

Date
04-05-2026

Investigator Name
Samad Afolabi

1. Case Summary

On December 9, 2009, a citizen named Aaron Greene bought a used computer on the website Craigslist from a seller named Terry Johnson. When Greene turned the computer on, she discovered that the previous owner had not wiped the hard drive. Greene found illegal "kitty" exploitation pictures and videos on the computer and called the Monterey Police Department.

The police took the computer (logged as evidence MT-2009-12-015-EV001). Forensic analysts looked at it and figured out the computer originally belonged to a local startup company called M57 Patents. M57 Patents does art patent searches and had only been open for about a month.

The computer was previously used by an M57 employee named Jo. The key people involved in this investigation at M57 Patents are:

- Pat McGoo: The CEO and owner of the company.
- Terry Johnson: The IT Administrator, who sold the computer.
- Jo Smith: A patent researcher and the alleged primary user of the computer that contained the illegal files.
- Charlie Brown: Another patent researcher.

Detectives met with the CEO, Pat McGoo, who agreed to let the police search the company's computers. The police also secured a legal search warrant (MT-2009-12-015-W001) specifically to seize any data and USB thumb drives belonging to Jo Smith. The police gathered forensic copies of four USB drives, network data, and four computer hard drives. We are now analyzing this data to find out who downloaded the illegal files and if any other crimes occurred.

1.1 Legal & Technical Justification

The initial discovery of the illegal files was made by a private citizen, Aaron Greene. Because she is not a government agent, the Fourth Amendment rules protecting against illegal search and seizure do not apply to her discovering the files. When the police went to M57 Patents, the CEO, Pat McGoo, gave them voluntary, third-party consent to search the business's electronic assets. To ensure the investigation was legally sound, the police also acquired a search warrant signed by a judge specifically targeting Jo Smith's digital items. This two-step process ensures all the digital evidence we analyze is legally acquired and admissible in court.

SIMULATED DETECTIVE REPORT – EDUCATIONAL USE ONLY

City of Monterey, CA Police Department
Detective Investigation Status Report

Date	09 December 2009
Time	11:30 AM
Affiant	Detective Joe Friday
Case Number	MT-2009-12-015
Report Number	001

Summary
Possible cat exploitation possession investigation.

Cause for Action
Received a call from: Aaron Greene 123 Cloudy Hill Drive Monterey, CA 93940 Aaron Greene had purchased a used computer from a seller on Craig’s List, a web site. When Aaron Greene started this computer, she found that the previous owner had not erased the hard disk computer drive. She claims to have found kitty exploitation movies and pictures on the computer.

Action Taken
Officers were sent to Aaron Greene’s house to take possession of the computer, which Greene voluntarily released. Greene also gave the name of the seller: Terry Johnson Email: No other information was available.

Outcome
Computer has been recovered, logged as evidence (MT-2009-12-015-EV001) and sent to forensics lab.

Detective Status Report documenting the initial seizure of the computer from Aaron Greene and chain of custody.

2. Digital Evidence Examined

To conduct a comprehensive investigation, we analyzed a specific set of digital artifacts acquired from M57 Patents. The evidence falls into four primary categories: hard drive images, USB drive images, random access memory (RAM) captures, and network traffic logs.

The collected evidence directory contains the following items:

- Hard Drive Images (.E01):
 - charlie-2009-12-11.E01
 - jo-2009-12-11-001.E01
 - jo-2009-12-11-002.E01
 - pat-2009-12-11.E01
 - terry-2009-12-11-001.E01
 - terry-2009-12-11-002.E01
- USB Drive Images (.E01):
 - charlie-work-usb-2009-12-11.E01
 - jo-favorites-usb-2009-12-11.E01
 - jo-work-usb-2009-12-11.E01
 - terry-work-usb-2009-12-11.E01
- RAM Images (.zip):
 - charlie-2009-12-11.mddramimage.zip
 - jo-2009-12-11.mddramimage.zip
 - pat-2009-12-11.mddramimage.zip
 - terry-2009-12-11.mddramimage.zip
- Network Traffic Captures (.pcap.gz): 49 packet capture files logging network activity from November 13, 2009, through December 13, 2009.

The hard disk images, RAM images, and USB drives belonging to Jo Smith were seized pursuant to a search warrant authorized by a judge. The evidence belonging to the other employees (Terry, Charlie, and Pat) were legally obtained via voluntary third-party consent from the company CEO, Pat McGoo, who agreed to release all electronic assets.

Technically, the physical storage devices (hard drives and USBs) were acquired using the .E01 (EnCase image) format. The .E01 files natively embed cryptographic hash values (like MD5 or SHA-1) and case metadata directly into the file. This makes .E01 highly reliable for court.

The RAM images capture the "volatile" memory—data that would be permanently lost if the computers were turned off. This is crucial for finding active processes, open network connections, or unencrypted passwords. The .pcap (packet capture) files represent a historical log of the data moving in and out of the company's network, allowing us to reconstruct web browsing, file transfers, and emails.

File Name	MD5 Hash
evidence/charlie-2009-12-11.E01	0377b3d41bbbc295a1c9f00aa07ee174
evidence/jo-2009-12-11-001.E01	a6c44b7387a67333b8566955dcad6f50
evidence/jo-2009-12-11-002.E01	434332cdbdb1463606cbea2d7a625745
evidence/pat-2009-12-11.E01	b30759487bb399387bbad6e776b1ed94
evidence/terry-2009-12-11-001.E01	e21dda28db6ef060c294964dda624265
evidence/terry-2009-12-11-002.E01	0008836bc52d4a4e717b43b09b558e58
evidence/usb/charlie-work-usb-2009-12-11.E01	9c0de6c8532d7a66ddcf01861dfb6535
evidence/usb/jo-favorites-usb-2009-12-11.E01	ad1d03cbdb8d81e918899479360f50dc
evidence/usb/jo-work-usb-2009-12-11.E01	8f23279deb398c3245829a98bc8fc1bd
evidence/usb/terry-work-usb-2009-12-11.E01	e07f26954b23db1a44dfd28ecd717da9

Table 1: Verified cryptographic hashes of the acquired .E01 files.

3. Forensic Tools and Methods

To ensure the integrity of the evidence and maintain a secure, reproducible analysis environment, the examination was conducted using Autopsy and a custom-built Linux container (Ubuntu 22.04 LTS). This container operated via Docker within a Windows Subsystem for Linux (WSL) host. The forensic toolset included a combination of command-line utilities for targeted extraction and graphical interfaces for comprehensive data visualization.

The primary tools utilized and their specific functions within the investigation include:

- The Sleuth Kit (TSK) and Autopsy: TSK was used within the Linux container for rapid, command-line based file system analysis and targeted data extraction. Also, Autopsy was used on the host Windows machine to visually browse directories, build timelines, and review recovered artifacts in a graphical interface.
- bulk_extractor: This tool was chosen to rapidly scan the raw disk images for patterns—such as email addresses—without needing to mount or parse the underlying file systems.
- ewf-tools: Used to safely mount and interact with the .E01 forensic disk images.

To prevent the accidental alteration or destruction of evidence, the original forensic images and network captures were mounted into the analysis container using a strict read-only (:ro) protocol. All subsequent analysis scripts and extracted results were directed to a separate, isolated directory.

This ensured that the source evidence remains untouched, establishing a clear and defensible chain of custody. The tools selected (such as The Sleuth Kit) are industry-standard, open-source applications that have been widely vetted and accepted within the scientific and legal communities, satisfying the requirements for reliable expert methodology in court.

4. Legal Authority

The legal foundation for this investigation is built upon three distinct events: a discovery by a private citizen, corporate consent from the business owner, and a targeted search warrant.

The primary crime under investigation is a felony violation of Section 311.3(a), Exploitation of a non-consenting entity. Based on the search warrant and affidavit, the investigation prioritized the analysis of the available images in the following order: First, the initial computer sold on Craigslist to verify the presence of illicit material. Second, the specific hard drives, RAM, and USB thumb drives assigned to or possessed by Jo Smith, as these were the primary target of the sworn affidavit. Finally, the broader corporate network data and the machines of the other employees (Terry, Charlie, and Pat) to provide context and identify any other illicit activities.

4.1 Legal Justification

To ensure all evidence is admissible in court, we have verified that no Fourth Amendment violations occurred regarding unlawful search and seizure. The acquisition of evidence in this case falls under three legally recognized parameters:

- The Private Search Exception: The initial illicit material was discovered by Aaron Greene, a private citizen who purchased the computer online. Because Greene was acting independently and not as an agent of the government, her discovery does not constitute a Fourth Amendment search. Law enforcement was legally permitted to accept and analyze the computer she voluntarily surrendered under the plain view doctrine.
- Third-Party Consent: To analyze the broader M57 Patents network and the devices used by Terry, Charlie, and Pat, the police acquired voluntary consent from the company's CEO, Pat McGoo. In a corporate environment, an employer generally retains the legal authority to consent to a search of company-owned hardware and networks. This consent effectively waives any reasonable expectation of privacy the employees might have claimed over the corporate evidence.
- The Warrant and Probable Cause: Because Jo Smith was the primary suspect associated with the Craigslist computer, law enforcement took the extra step of securing a search warrant specifically for Jo's items. Detective Joe Friday filed an affidavit outlining the facts of the Craigslist computer to establish probable cause. The resulting warrant (MT-2009-12-015-W001) satisfied the Fourth Amendment's "particularity requirement" by explicitly authorizing the seizure of usernames, RAM images, hard disk drives, and all USB thumb drives connected to Jo Smith between November 13 and December 12, 2009.

FOR THE FOLLOWING PROPERTY:

- 1. The subscriber information, usernames and account information for any and all computers assigned to Jo Smith from November 13, 2009 to December 12, 2009.
- 2. A RAM image for any and all computers assigned to Jo Smith from November 13, 2009 to December 12, 2009.
- 3. A hard disk drive image for all computers assigned to Jo Smith from November 13, 2009 to December 12, 2009.
- 4. All USB thumb drives in Jo Smith's possession, regardless of ownership.

FURTHER ORDERS

It is further ordered that the records obtained pursuant to this warrant (with the exclusion of prohibited/illegal material) may be shared with M57 Patents and the civil division of the Office of the Attorney General for purposes of any civil action related to the unauthorized use of corporate equipment, intellectual property, or work product.

The court orders that any items seized during the lawful service of this search warrant be disposed in accordance with the law, by the Monterey County Prosecutor, upon adjudication of the case. The officers serving the warrant are not authorized to return seized items to any known victim(s), as the items may contain illegal/prohibited material. Further application for such returns shall be issued by this court on evidence that the items contain no illegal/prohibited material, or that any such material has been destroyed, and also that the item itself serves no further material purpose in the investigation of illegal criminal or civil activity.

If necessary, searching officers are authorized to employ the use of outside experts, acting under the direct control of the investigating officers, to access and preserve any computer data.

AND TO SEIZE IT IF FOUND and bring forthwith before me, or this court, at he courthouse of this court. The Search Warrant and incorporated Affidavit was sworn as true and subscribed before me the ____ day of _____, 2009, at _____ A.M/P.M. Wherefore, I find probable cause for the issuance of this Search Warrant and do issue it.

(Signature of Magistrate)

Night search approved: YES ☐ NO ☐
(Magistrate's initials)

Executed By: _____ Date: _____ Time: _____

5. Investigative Findings

5.2 A number of professional contacts and outside persons (friends of the employees) may appear in this scenario. Who are they? Are they involved in any of the activities uncovered?

Analysis of the M57 corporate communications reveals a network of external contacts interacting with the employees. These individuals are categorized into active co-conspirators, unwitting purchasers of embezzled hardware, personal acquaintances serving as material witnesses to motive or sudden wealth, and innocent professional contacts. The digital evidence directly implicates specific outside actors in purchasing stolen intellectual property and participating in extortion schemes orchestrated by M57 personnel, while completely clearing other contacts of criminal involvement.

The external contacts Jamie and Andy are implicated as active co-conspirators and buyers in intellectual property theft, elevating Charlie Brown's actions from policy violations to felony corporate espionage. In addition, Aaron Greene and Jean Sizemore are cleared as innocent third-party purchasers; their documented interactions legally establish the chain of custody for the embezzled hardware containing the illicit materials (California Penal Code 311.3(a)). Finally, the personal contacts act as digital character witnesses. Their correspondence provides highly admissible, corroborating evidence for the financial motives driving Terry Johnson's embezzlement.

Artifact Group 1: Active Co-Conspirators

- Jamie / "J" (jamie@project2400.com / jaime@project2400.com): Email correspondence extracted from Charlie Brown's workstation details a \$50,000 ("\$50 large") offer for sensitive trade secrets regarding a competitor (Nitroba). Charlie's outbound emails include the password "nitro," confirm the transmission of data via steganography, and contain explicit instructions to "delete these emails" to obscure the transaction.
- Andy (andy@swexpert.com): Email correspondence reveals Charlie Brown leveraging prior art to threaten the validity of Andy's "immortality patent." Subsequent emails show Charlie transmitting a "promised" picture (utilized as a steganographic carrier) accompanied by the directive to "Make sure you delete this."

Artifact Group 2: Hardware Buyers

- Aaron Greene (aarongreene12@gmail.com): Contact files and email records confirm relations with Terry Johnson for the \$1,000 purchase of the misappropriated Dell Computer, establishing Greene as an innocent purchaser and the initial whistle-blower.
- Jean Sizemore (jeansizemore3@gmail.com): Email correspondence in Terry Johnson's mailbox confirms negotiations for the \$200 purchase of an embezzled HP Printer and Dell Monitor.

Artifact Group 3: Material Witnesses

- Alix Pery & Rubin Fritz (alix.pery@yahoo.com, rubinfritz31@mail.com): Email correspondence demonstrates Charlie Brown boasting of a sudden, unexplained influx of cash. The communications detail plans for a Mediterranean cruise and the purchase of high-end vehicles (Ford Shelby GT500, Ferraris).
- "Cod" & Jesse (ghost.wisp@live.com, tech.norat@hotmail.com): Email correspondence demonstrates Terry Johnson's financial desperation and unregulated gambling habits. Terry explicitly informs Cod of his financial struggles and his urgent need to "get cash tonight" for \$50 buy-in poker tournaments.

5.3 Is Terry involved in anything illicit or against policy? If so, describe in detail what he did if anything.

Forensic analysis of the AppCompatCache (Application Compatibility Cache) on Terry Johnson's workstation reveals a systematic and unauthorized deployment of clandestine surveillance and data spoliation tools. On December 3, 2009, the user actively downloaded, installed, and executed multiple commercial and underground keylogger applications designed to secretly record user keystrokes. Furthermore, Terry deployed multiple secure data-wiping utilities (CCleaner and Eraser). This combination of software indicates an illicit effort to establish covert monitoring capabilities over the network while possessing the tools necessary to permanently destroy the digital evidence of these activities.

The execution of covert surveillance tools (keyloggers) on a corporate system represents a critical breach of acceptable use policies and potentially violates federal and state wiretap/electronic privacy statutes. Even with IT administrative privileges, deploying unauthorized software designed to silently intercept passwords and communications without user consent constitutes illicit monitoring. Additionally, the subsequent installation of the Eraser wiping utility on December 10, 2009 (one day after the police investigation into the Craigslist computer commenced), strongly indicates intentional data spoliation and a deliberate attempt to obstruct forensic discovery.

Artifacts from SYSTEM Registry

- Location: /img_terry-2009-12-11-001.E01/vol_vol2/Windows/System32/config/SYSTEM
- MD5 Hash: 796801a9f94d142a488762e6d321cae9

Artifact Group 1: Keyloggers

Registry analysis confirms the execution of multiple keystroke logging applications in rapid succession on December 3, 2009:

- Target File: C:\Users\terry\Documents\Downloads\skl0g\skl0g.exe (Executed: 2009-12-03 17:17:21).
- Target File: C:\Users\terry\Documents\Downloads\xpadvancedkeylogger.exe (Executed: 2009-12-03 17:30:39).
- Target File: C:\Program Files\XP Advanced Keylogger\unins000.exe (Executed: 2009-12-03 17:30:56). (*Indicates installation, testing, and rapid uninstallation*).
- Target File: C:\Users\terry\Documents\Downloads\keylogger\FamilyKeyLogger-setup.exe (Executed: 2009-12-03 17:33:11).

Artifact Group 2: Anti-Forensics

The user actively executed applications designed to permanently remove data and cover digital tracks:

- Target File: C:\Program Files\CCleaner\CCleaner.exe (Executed: 2009-11-24 21:49:10).
- Target File: C:\Users\terry\Documents\Downloads\Eraser-5.8.7_setup.exe (Executed: 2009-12-10 16:27:43).

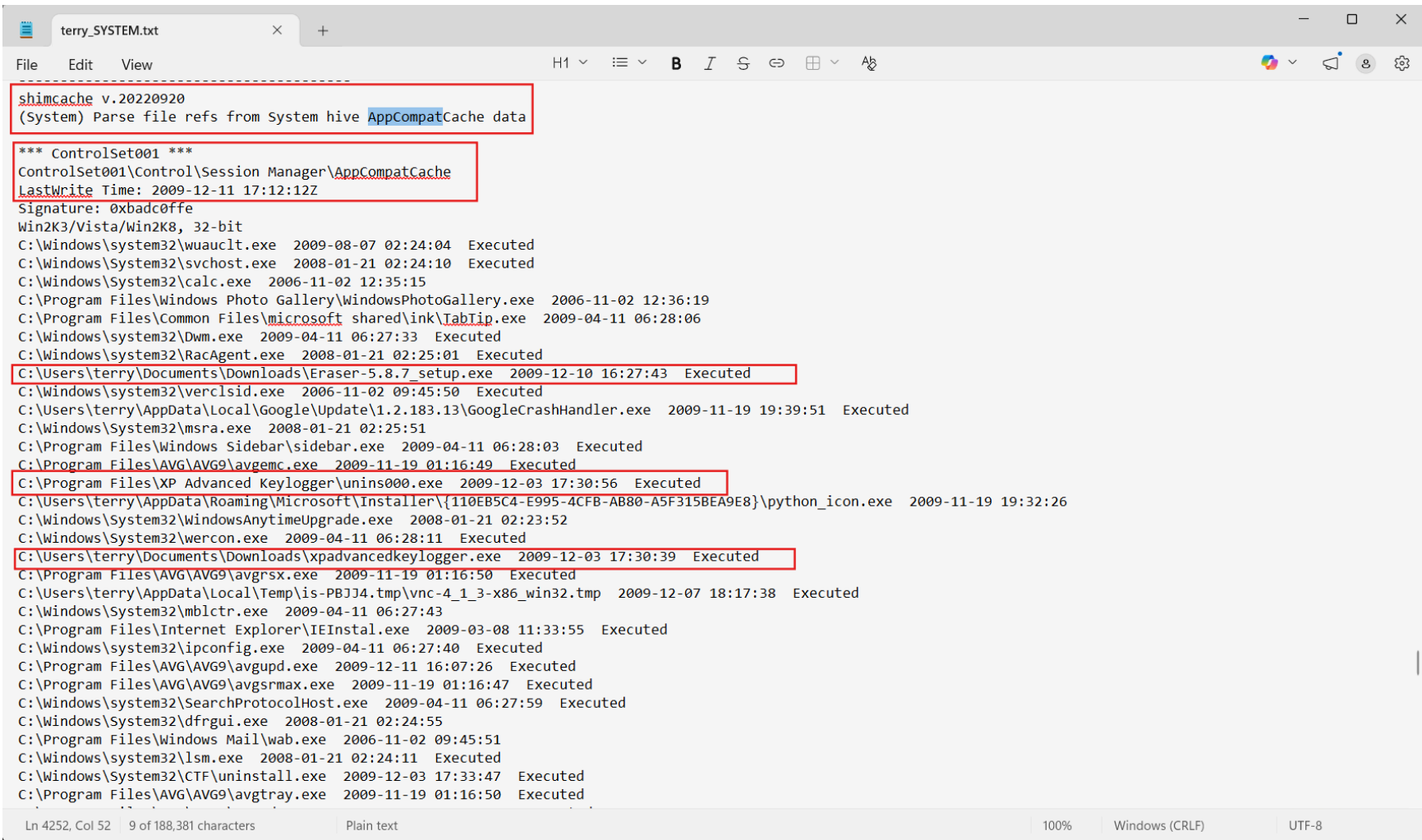


Exhibit A: Parsed AppCompatCache from the terry_SYSTEM hive.

5.4 Is Jo the owner of the files you found from Jo’s USB drive or any drive that is evidence of Jo? What evidence is there to confirm or refute this?

Cryptographic comparison of file artifacts establishes a definitive link between Jo’s M57 workstation (assigned to the user account “jo”) and files located on multiple USB storage devices. Identical digital fingerprints were verified for a document found both on the user’s desktop and the “work” USB drive, confirming Jo’s possession and control of those files. A similar correlation was identified for a multimedia file present on the “favorites” USB drive and within the workstation’s \$OrphanFiles directory. Its presence in a deleted state on the local hard drive indicates prior possession on the workstation, followed by a deliberate attempt to remove the data.

This evidence falls squarely within the authorized scope of the executed search warrant (MT-2009-12-015-W001), which mandates the seizure and forensic analysis of the hard disk drive assigned to Jo Smith, as well as all USB thumb drives in Jo Smith’s possession, regardless of ownership. The cryptographic hash matches satisfy the evidentiary burden of proving possession by directly linking the material to the user’s operating environment. Furthermore, the recovery of the multimedia file as an orphaned (deleted) artifact is consistent with alleged violations of California Penal Code 311.3(a) and supports an inference of consciousness of guilt through attempted evidence destruction.

Artifact Pairing 1: The "Work" USB

Workstation Path: ` /img\_jo-2009-12-11-001.E01/vol\_vol2/Documents and Settings/Jo/Desktop/Papers/Papers1/10728.MeshNetworks.Nancy+Baker.pdf`

USB Path:` /img\_jo-work-usb-2009-12-11.E01/vol\_vol2/Papers1/10728.MeshNetworks.Nancy+Baker.pdf`

MD5 Hash: `27e6852d221ae358ab1683c226a52828`

Artifact Pairing 2: The "Favorites" USB

Workstation Path:` /img\_jo-2009-12-11-001.E01/vol\_vol2/\$OrphanFiles/Videos/KittyMontage.mov`

USB Path: ` /img\_jo-favorites-usb-2009-12-11.E01/vol\_vol2/Videos/KittyMontage.mov`

MD5 Hash: `7d4058485071e25d482fd18fb2ac58bd`

Relevant Metadata: The workstation file exists in an "Unallocated" state.

CaseViewToolsWindowHelp

Add Data Source

Images/Videos

Communications

Geolocation

Timeline

Discovery

Generate Report

Close Case

Listing

TableThumbnailSummary

104 Results

Page: 1 of 1

Pages:

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Save Table as CSV

Name	S	C	O	Size	Flags(Dir)	MD5 Hash	Modified Time	Created Time	Change Time
10728.MeshNetworks.Nancy+Baker.pdf			1	81255	Allocated	27e6852d221ae358ab1683c226a52828	2009-11-17 19:46:18 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
1085.Architecture.Zelma+Reigle.pdf			1	55068	Allocated	f9cfd84a938329efec396af5c34b2c7f	2009-11-17 20:42:32 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
11384.LocationInDslpit.Ronald+Clark.pdf			1	49884	Allocated	d8ed0b9c80f68acaba560253eb57f3e2	2009-11-17 17:49:26 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
11726.transistor.Paul+Jackson.pdf			1	49392	Allocated	771cdc5adddb78ee9c5e2e89fa2f65e6	2009-11-17 17:41:26 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
12097.RPCs.Fernando+Loring.pdf			1	77349	Allocated	8a4964b39ef4dcacc6eef5d12619e1b2	2009-11-17 20:09:52 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
12543.UNIVAC.Brasil+Gerraro.pdf			1	45127	Allocated	6b9d939d24321bdd54ff7fb4c9a13dd9	2009-11-17 20:27:08 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
13030.Smalltalk.Hugh+Brinkmann.pdf			1	134109	Allocated	b27061baeeafa39eb98860b216eb4ddb	2009-11-17 20:29:20 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
13136.OundyVakeelJavier+Sloop.pdf			1	63975	Allocated	082f33d5987c6f42431635b5c9fc7f4f	2009-11-17 20:08:18 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
15238.IlkHob.Neil+Speno.pdf			1	106421	Allocated	f0883b197b64a0bb5b18b361ccfd4f4	2009-11-17 20:23:24 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
16860.BTrees.Lance+Fason.pdf			1	131866	Allocated	86b549775347befd33ccacb5fe8eacd0	2009-11-17 20:12:16 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
17396.SMPs.Noemi+Yule.pdf			1	47194	Allocated	39f14b73c32f6a4ff7c0965ea4b488e3	2009-11-17 20:43:56 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
21427.BolnWhang.Deborah+Collins.pdf			1	55146	Allocated	ca91480744ba5082ff870a2f0ea6289b	2009-11-17 20:00:54 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
22231.PineySirdar.Emilia+Wyer.pdf			1	43311	Allocated	9234fe1f7eb78b3ba63c1966451cf268	2009-11-17 20:39:34 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
2300.FaneMesel.Sharon+Phillips.pdf			1	92322	Allocated	fab36beecd38eb9d19a4e7eb0394df1	2009-11-17 19:56:32 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST
27172.Wanger.Kelly+Koepp.pdf			1	89642	Allocated	1e7685b217b3198d9e6ac8f7462db2d1	2009-11-17 20:41:52 EST	2009-12-10 17:41:52 EST	2009-12-10 17:41:52 EST

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

Metadata

Name:

/img_jo-2009-12-11-001.E01/vol_vol2/Documents and Settings/Jo/Desktop/Papers/Papers1/10728.MeshNetworks.Nancy+Baker.pdf

Type:

File System

MIME Type:

application/pdf

Size:

81255

File Name Allocation:

Allocated

Metadata Allocation:

Allocated

Modified:

2009-11-17 19:46:18 EST

Accessed:

2009-12-10 17:41:52 EST

Created:

2009-12-10 17:41:52 EST

Changed:

2009-12-10 17:41:52 EST

MD5:

27e6852d221ae358ab1683c226a52828

SHA-256:

e6155fa1dd7fc8b772586ecff2432902fe15cd10f29648ff4000b3c85c4eebdf

Hash Lookup Results:

UNKNOWN

CaseViewToolsWindowHelp

Add Data Source

Images/Videos

Communications

Geolocation

Timeline

Discovery

Generate Report

Close Case

Listing

TableThumbnailSummary

96 Results

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12097.RPCs.Fernando+Loring.pdf			1	77349	Allocated	8a4964b39ef4dcacc6eef5d12619e1b2	2009-11-17 17:09:52 EST	2009-12-10 14:30:46 EST	0000-00-00 00:00:00
12543.UNIVAC.Brasil+Gerraro.pdf			1	45127	Allocated	6b9d939d24321bdd54ff7fb4c9a13dd9	2009-11-17 17:27:08 EST	2009-12-10 14:30:46 EST	0000-00-00 00:00:00
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13136.OundyVakeelJavier+Sloop.pdf			1	63975	Allocated	082f33d5987c6f42431635b5c9fc7f4f	2009-11-17 17:08:18 EST	2009-12-10 14:30:46 EST	0000-00-00 00:00:00
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16860.BTrees.Lance+Fason.pdf			1	131866	Allocated	86b549775347befd33ccacb5fe8eacd0	2009-11-17 17:12:16 EST	2009-12-10 14:30:46 EST	0000-00-00 00:00:00
17396.SMPs.Noemi+Yule.pdf			1	47194	Allocated	39f14b73c32f6a4ff7c0965ea4b488e3	2009-11-17 17:43:56 EST	2009-12-10 14:30:47 EST	0000-00-00 00:00:00
21427.BolnWhang.Deborah+Collins.pdf			1	55146	Allocated	ca91480744ba5082ff870a2f0ea6289b	2009-11-17 17:00:54 EST	2009-12-10 14:30:47 EST	0000-00-00 00:00:00
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27172.Wanger.Kelly+Koepp.pdf			1	89642	Allocated	1e7685b217b3198d9e6ac8f7462db2d1	2009-11-17 17:41:52 EST	2009-12-10 14:30:47 EST	0000-00-00 00:00:00

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

Metadata

Name:

/img_jo-work-usb-2009-12-11.E01/vol_vol2/Papers1/10728.MeshNetworks.Nancy+Baker.pdf

Type:

File System

MIME Type:

application/pdf

Size:

81255

File Name Allocation:

Allocated

Metadata Allocation:

Allocated

Modified:

2009-11-17 16:46:18 EST

Accessed:

2009-12-10 00:00:00 EST

Created:

2009-12-10 14:30:45 EST

Changed:

0000-00-00 00:00:00

MD5:

27e6852d221ae358ab1683c226a52828

SHA-256:

e6155fa1dd7fc8b772586ecff2432902fe15cd10f29648ff4000b3c85c4eebdf

Hash Lookup Results:

UNKNOWN

Exhibit A: Screenshots showing the Metadata tab for `10728.MeshNetworks.Nancy+Baker.pdf` on both the local desktop and the "work" USB drive. The matching MD5 and SHA-256 hashes are highlighted to prove identical file composition.

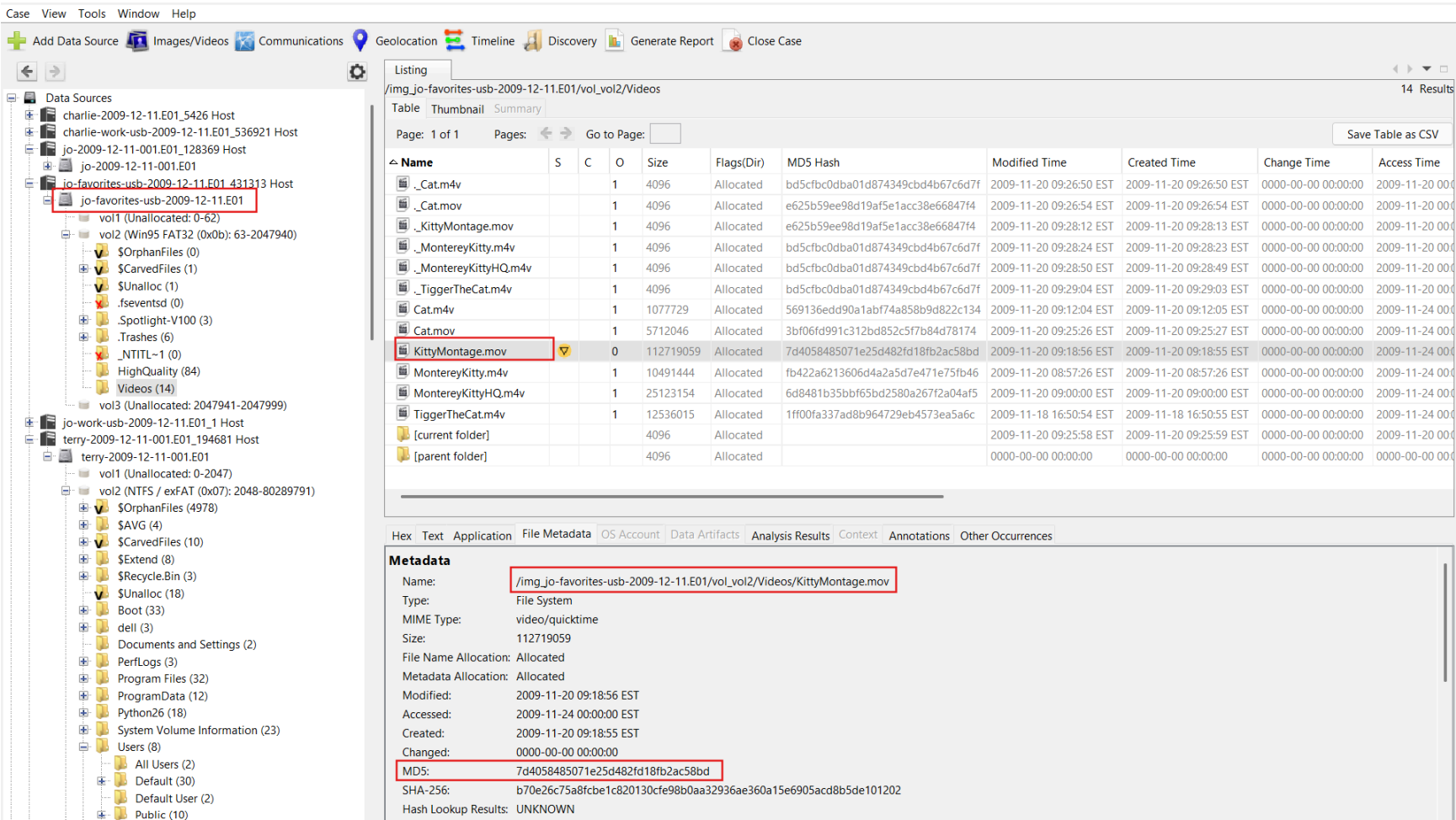
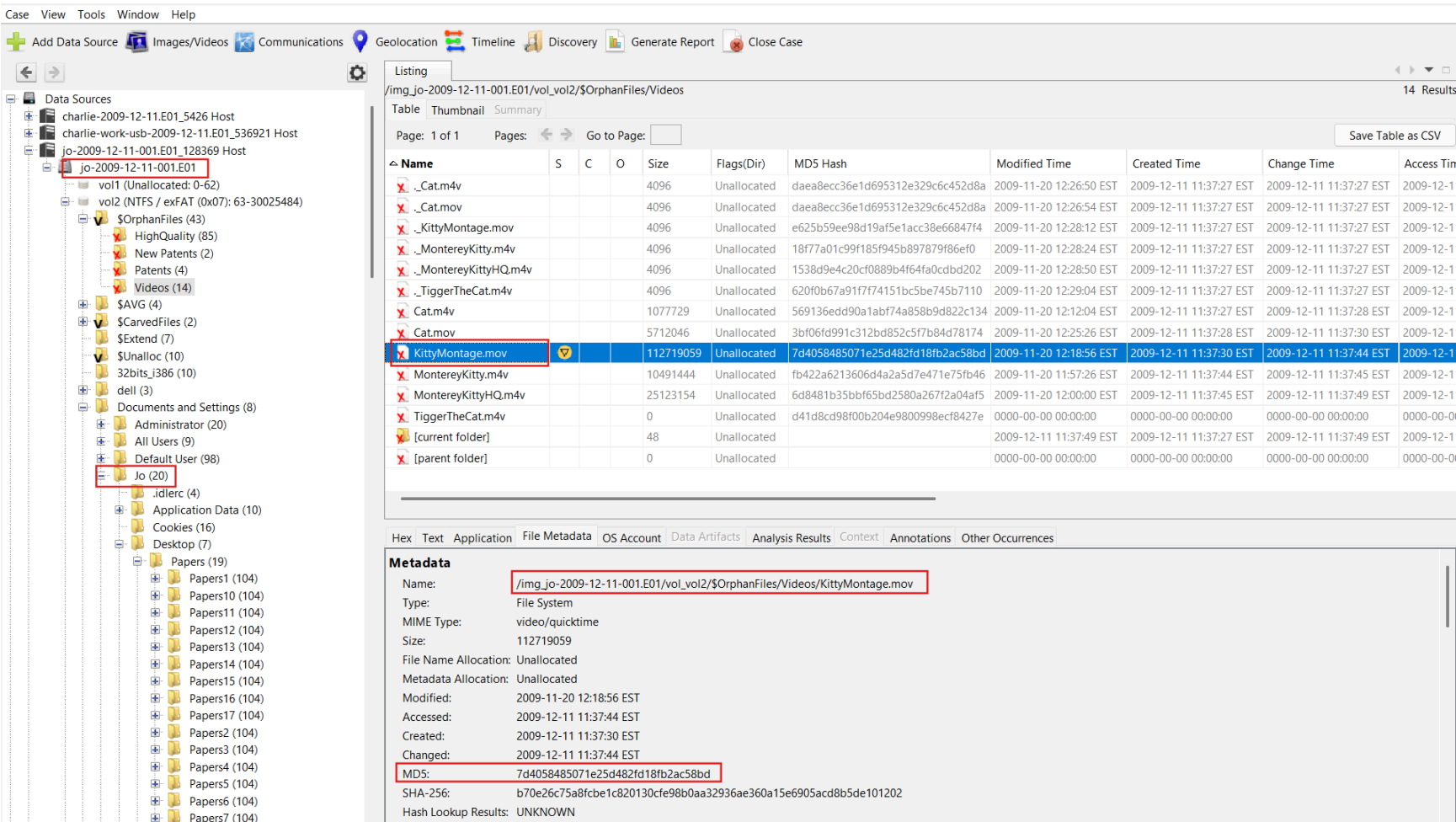


Exhibit B: Screenshot demonstrating the Metadata tab for `KittyMontage.mov`. The matching MD5 and SHA-256 hashes are highlighted to prove identical file composition.

5.5 Terry has attempted to eliminate evidence of criminal activity by deleting old emails. How did he do this? Was it successful? Can emails be recovered from the drive image in police evidence?

Analysis of Terry Johnson's workstation reveals a failed attempt to eliminate evidence of his activities. The method of data destruction was limited to "soft-deleting" messages within the Windows Mail client application, which moved incriminating emails into the Deleted Items folder without permanently removing them or using third-party disk-wiping utilities. Also, the evidence destruction attempt was entirely unsuccessful: the deleted emails remain fully allocated on the file system and are completely recoverable in plain text format. Notably, the timestamp of this deletion activity (December 9, 2009) aligns precisely with the date law enforcement was notified of the illicit computer sale, indicating a panicked and reactive attempt to conceal involvement.

The examination of Terry Johnson's email repository is legally authorized under the voluntary consent provided by M57 Patents CEO Pat McGoo, who released all corporate electronic assets for forensic review.

- Deletion Method: Soft-deletion via Windows Mail (moved to trash bin). No third-party wiping utilities were identified.
- Artifact Location: /img_terry-2009-12-11-001.E01/vol_vol2/Users/terry/AppData/Local/Microsoft/Windows Mail/Local Folders/Deleted Items/
- File Status: "Allocated" (Fully intact and recoverable).
- Example Recovered File: 1CFA4233-0000001E.eml
 - MD5: 4c24e5f93d1971f8014949b60a892e63
 - Subject: "Your craigslist posting 'Dell Computer For Sale'"
 - Deletion/Modified Timestamp: 2009-12-09 17:04:54 EST (Correlates with the initial police report).

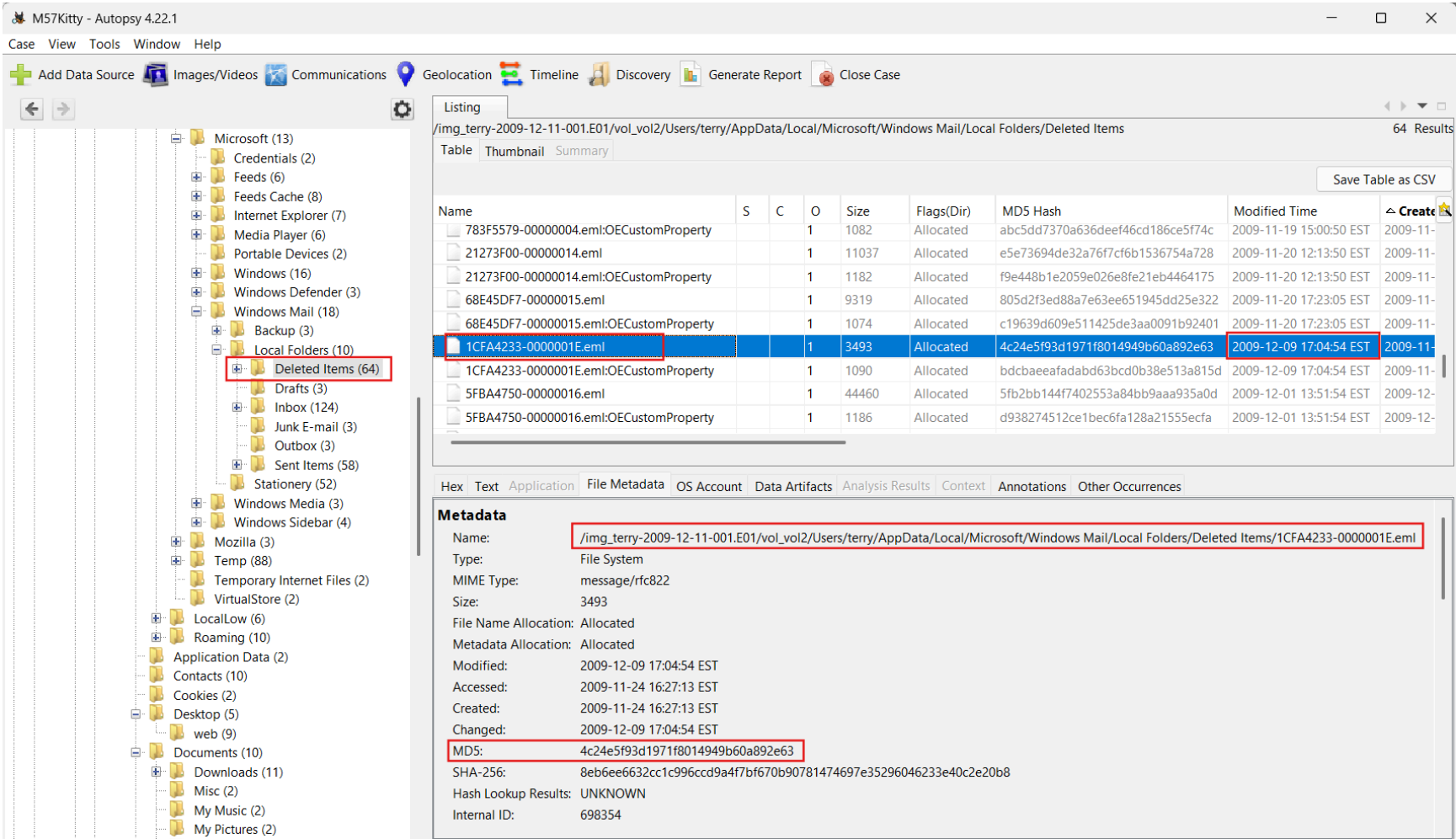


Exhibit A: Screenshot displaying the contents of Terry's Deleted Items folder and example recovered file.

5.6 Charlie sends a number of emails with attachments that have been protected or otherwise obfuscated. What are they? Can the original information be recovered?

Analysis of Charlie's sent emails to “jamie@project2400.com” and “andy@swexpert.com” reveals the transmission of obfuscated attachments designed to conceal intellectual property. The obfuscation was three-pronged:

- Encryption: Two target files were encrypted within a password-protected ZIP archive (01.zip).
- Plaintext Steganography: The decryption key (password=immortal) was embedded directly into the hexadecimal padding of a decoy image (microscope1.jpg).
- Payload Injection: A compressed data stream was injected into the COM (Comment) block of a second decoy image (astronaut1.jpg).

By extracting this embedded key, I was able to successfully decrypt the ZIP archive, recovering the original payload consisting of two distinct TIFF image files detailing US Patents 6,982,168 and 5,026,637.

The acquisition and analysis of these email communications falls legally within the authorized scope of the investigation, as M57 Patents CEO Pat McGoo voluntarily agreed to release all electronic assets from the company in support of this investigation. The forensic extraction of the password from the image file's hex code constitutes a lawful analysis of natively seized data. The extraction of the injected data stream within the second image produced an intact OpenDocument Text (.odt) file.

Artifacts (Email Attachments)

- Obfuscated file: /img_charlie-2009-12-11.E01/vol_vol2/Documents and Settings/Charlie/Application Data/Thunderbird/Profiles/4zy34x9h.default/Mail/Local Folders/Sent/01.zip
 - MD5: 4fa239c22e5fb7b934a1bf68e4e0e2e7
- Steganographic Key Carrier: /img_charlie-2009-12-11.E01/vol_vol2/Documents and Settings/Charlie/Application Data/Thunderbird/Profiles/4zy34x9h.default/Mail/Local Folders/Sent/microscope1.jpg
 - MD5: 4be2c4abb48c4389ca798e6c21736ea1
 - Key Location: Hex offset 0x00000080 containing ASCII string password=immortal.

- File Carrier: /img_charlie-2009-12-11.E01/vol_vol2/Documents and Settings/Charlie/Application Data/Thunderbird/Profiles/4zy34x9h.default/Mail/Local Folders/Sent/astronaut1.jpg
 - MD5: 45eade24b3a89b21fed303310ccbd54

Recovered Artifacts (Extracted from 01.zip)

- Recovered Document 1: us006982168-001.tif
 - MIME Type: image/tiff (Verified by 49 49 2a 00 file signature)
 - MD5: c412dfcd8bca1333d9356d710dd7a01a
- Recovered Document 2: us005026637-001.tif
 - MIME Type: image/tiff (Verified by 49 49 2a 00 file signature)
 - MD5: b526e6c7a244d62e7120f3f804d76d8d

Recovered Artifacts (Extracted from astronaut1.jpg)

- Raw Zlib Stream: AC.zlib
 - MD5: 7a038931698f78f2be2b0436fd9478ff
- Decoded Document: AC (Identified via file signature as OpenDocument Text)
 - MD5: 56fd56fc40b7c6d7b7572711f863bc8d

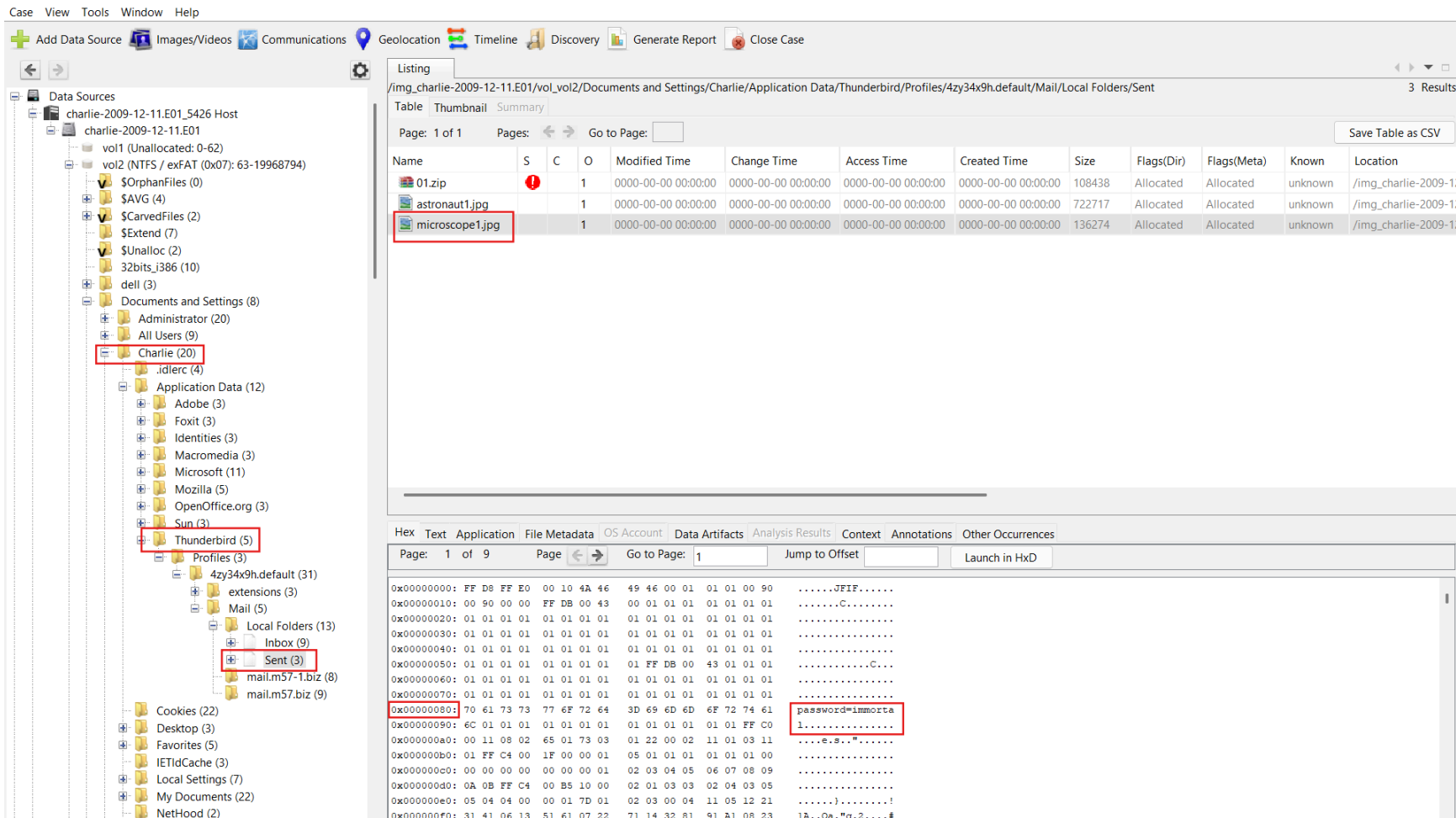


Exhibit A: Screenshot of the Autopsy Hex Viewer for microscope1.jpg showing the hidden password.

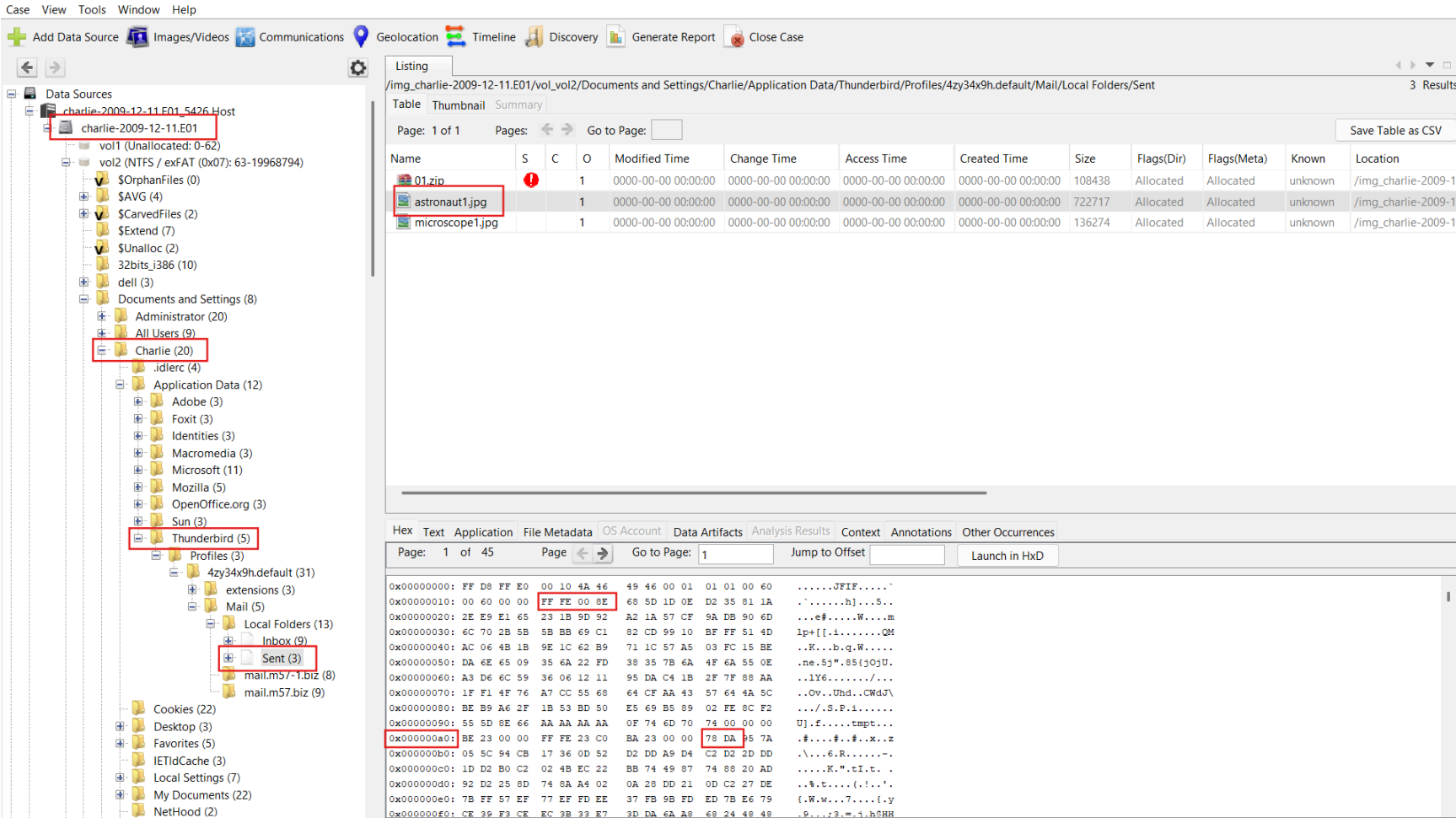


Exhibit B: Screenshot screenshot of the Autopsy Hex Viewer for astronaut1.jpg highlighting hex values proving the existence of an hidden artifact.

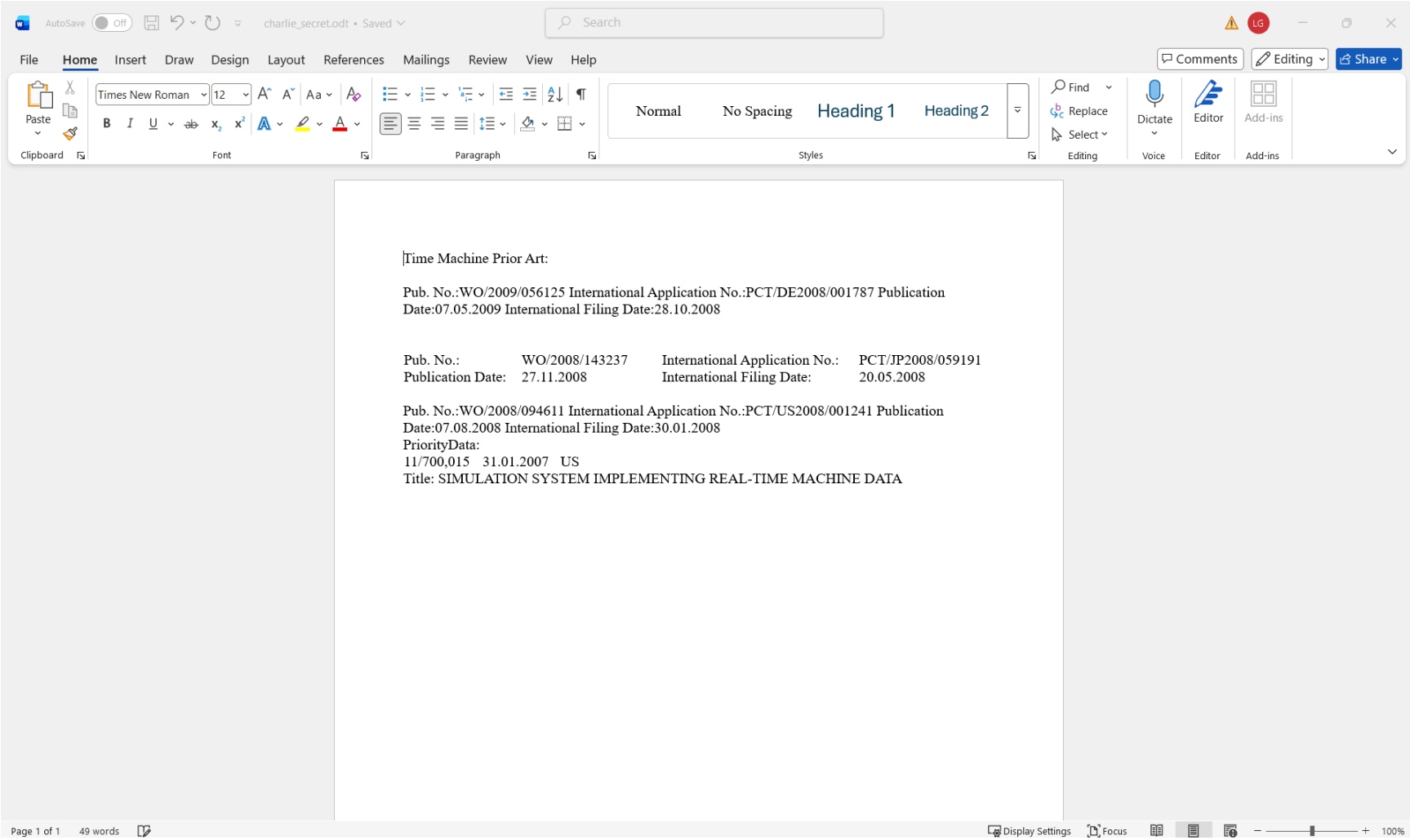


Exhibit C: Screenshot displaying the contents of the extracted artifact from atronaut1.jpg file.

5.7 A large number of processes are running on a typical Windows machine. However, the machines used by the m57 employees are not identical. Were there unusual or noteworthy processes running on any of the machines during the police capture?

Analysis of the live memory (RAM) captured across the four M57 corporate workstations shows a highly diverse software environment. The most eye-catching operational anomaly was found on Terry Johnson's workstation, where a secure data-wiping utility (Eraser.exe) was actively running as a hidden background process, indicating ongoing evidence destruction at the time of seizure while the workstations assigned to Jo, Charlie, and Pat showed no active encryption or wiping utilities in memory. Finally, the process trees confirm non-standardized corporate IT policies, as shown by employees running completely different email clients

(Thunderbird vs. Outlook Express). The analysis of volatile memory is legally authorized under the voluntary search consent provided by M57 Patents CEO Pat McGoo.

Evidence destruction (Terry)

- Process: Eraser.exe (PID: 596)
- Execution: "C:\Program Files\Eraser\Eraser.exe" -hide

Non-Standard Applications

- Charlie: thunderbird.exe (PID: 188)
- Pat: msimn.exe / Outlook Express (PID: 3020)

```
(venv) root@f8123716479d:/forensics# vol -f evidence/ram/terry-2009-12-11.winddramimage windows.pstree
Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime Au
dit Cmd Path
4 0 System 0x83076020 102 1488 N/A False 2009-12-11 19:04:29.000000 UTC N/A - --
* 408 4 smss.exe 0x852f0698 4 28 N/A False 2009-12-11 19:04:29.000000 UTC N/A \D
evice\HarddiskVolume1\Windows\System32\smss.exe \SystemRoot\System32\smss.exe \SystemRoot\System32\smss.exe
476 464 csrss.exe 0x85330bf8 11 586 0 False 2009-12-11 19:04:32.000000 UTC N/A \D
evice\HarddiskVolume1\Windows\System32\csrss.exe C:\Windows\system32\csrss.exe ObjectDirectory=\Windows SharedSecti
on=1024,12288,512 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 Ser
verDll=winsrv:ConServerDllInitialization,2 ProfileControl=Off MaxRequestThreads=16 C:\Windows\system32\csrss.exe
520 464 wininit.exe 0x852d8b70 3 99 0 False 2009-12-11 19:04:32.000000 UTC N/A \D
evice\HarddiskVolume1\Windows\System32\wininit.exe wininit.exe C:\Windows\system32\wininit.exe
* 576 520 avgrsx.exe 0x8539b020 28 273 0 False 2009-12-11 19:04:34.000000 UTC N/A \D
evice\HarddiskVolume1\Program Files\AVG\AVG9\avgrsx.exe "C:\Program Files\AVG\AVG9\avgrsx.exe" C:\Program Files\A
VG\AVG9\avgrsx.exe
** 692 576 avgcsrvx.exe 0x85419d90 8 182 0 False 2009-12-11 19:04:34.000000 UTC N/A \D
evice\HarddiskVolume1\Program Files\AVG\AVG9\avgcsrvx.exe /pipeName=99974200-7924-40f9-a51b-779a793952a6 /coreSdkOp
tions=30 /logConfFile="C:\ProgramData\avg9\temp\ee20105a-1938-46d7-952b-e76ab19c07f3-240-oopp.tmp" /loggerName=AVG.RS.Cor
e /binaryPath="C:\Program Files\AVG\AVG9\" /tempPath="C:\ProgramData\avg9\temp\" C:\Program Files\AVG\AVG9\avgcsrvx.e
xe
* 644 520 lsass.exe 0x853e9d90 9 552 0 False 2009-12-11 19:04:34.000000 UTC N/A \D
evice\HarddiskVolume1\Windows\System32\lsass.exe C:\Windows\system32\lsass.exe C:\Windows\system32\lsass.exe
* 652 520 lsm.exe 0x853f42f8 9 155 0 False 2009-12-11 19:04:34.000000 UTC N/A \Device\H
arddiskVolume1\Windows\System32\lsm.exe C:\Windows\system32\lsm.exe C:\Windows\system32\lsm.exe
* 628 520 services.exe 0x853f4d90 6 221 0 False 2009-12-11 19:04:34.000000 UTC N/A \D
evice\HarddiskVolume1\Windows\System32\services.exe C:\Windows\system32\services.exe C:\Windows\system32\servic
es.exe
** 480 628 svchost.exe 0x85781ce8 5 120 0 False 2009-12-11 19:04:52.000000 UTC N/A \D
evice\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\system32\svchost.exe -k NetworkServiceNetworkRestricted
C:\Windows\system32\svchost.exe
** 1284 628 svchost.exe 0x849dac78 37 557 0 False 2009-12-11 19:04:39.000000 UTC N/A \D
evice\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestrictedC:\
Windows\System32\svchost.exe
*** 156 1284 dwm.exe 0x83b8e020 3 74 1 False 2009-12-11 19:21:21.000000 UTC N/A \Device\H
arddiskVolume1\Windows\System32\dwm.exe "C:\Windows\system32\Dwm.exe" C:\Windows\system32\Dwm.exe

528 512 csrss.exe 0x852dbd90 9 256 1 False 2009-12-11 19:04:32.000000 UTC N/A \D
evice\HarddiskVolume1\Windows\System32\csrss.exe C:\Windows\system32\csrss.exe ObjectDirectory=\Windows SharedSecti
on=1024,12288,512 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 Ser
verDll=winsrv:ConServerDllInitialization,2 ProfileControl=Off MaxRequestThreads=16 C:\Windows\system32\csrss.exe
564 512 winlogon.exe 0x853676c8 3 125 1 False 2009-12-11 19:04:34.000000 UTC N/A \D
evice\HarddiskVolume1\Windows\System32\winlogon.exe winlogon.exe C:\Windows\system32\winlogon.exe
2840 1480 explorer.exe 0x85636d90 24 564 1 False 2009-12-11 19:21:21.000000 UTC N/A \D
evice\HarddiskVolume1\Windows\explorer.exe C:\Windows\Explorer.EXE C:\Windows\Explorer.EXE
* 1120 2840 jusched.exe 0x83d39020 1 53 1 False 2009-12-11 19:21:23.000000 UTC N/A \D
evice\HarddiskVolume1\Program Files\Java\jre6\bin\jusched.exe "C:\Program Files\Java\jre6\bin\jusched.exe" C:\Program
Files\Java\jre6\bin\jusched.exe
* 2792 2840 avgtray.exe 0x83cd8330 10 223 1 False 2009-12-11 19:21:23.000000 UTC N/A \D
evice\HarddiskVolume1\Program Files\AVG\AVG9\avgtray.exe "C:\Program Files\AVG\AVG9\avgtray.exe" C:\Program
Files\AVG\AVG9\avgtray.exe
* 4012 2840 cmd.exe 0x83bd9598 1 22 1 False 2009-12-11 19:51:25.000000 UTC N/A \Device\H
arddiskVolume1\Windows\System32\cmd.exe "C:\Windows\System32\cmd.exe" C:\Windows\System32\cmd.exe
** 2560 4012 win32dd1.exe 0x83951c18 1 23 1 False 2009-12-11 20:00:12.000000 UTC N/A \D
evice\HarddiskVolume2\RAM\windd\32bits_i386\win32dd1.exe win32dd1.exe /d /f E:\RAM\terry-2009-12-11-002.dmp E:\
RAM\windd\32bits_i386\win32dd1.exe
* 2932 2840 sidebar.exe 0x83d32580 13 393 1 False 2009-12-11 19:21:23.000000 UTC N/A \D
evice\HarddiskVolume1\Program Files\Windows Sidebar\sidebar.exe "C:\Program Files\Windows Sidebar\sidebar.exe" /au
toRun C:\Program Files\Windows Sidebar\sidebar.exe
* 596 2840 Eraser.exe 0x83cd49c0 2 122 1 False 2009-12-11 19:21:26.000000 UTC N/A \D
evice\HarddiskVolume1\Program Files\Eraser\Eraser.exe "C:\Program Files\Eraser\Eraser.exe" -hide C:\Program Files\E
raser\Eraser.exe
* 1752 2840 BrStsWnd.exe 0x83d36a98 3 81 1 False 2009-12-11 19:21:23.000000 UTC N/A \D
evice\HarddiskVolume1\Program Files\Brownie\BrStsWnd.exe "C:\Program Files\Brownie\BrStsWnd.exe" Autorun C:\Program
Files\Brownie\BrStsWnd.exe
** 3136 1752 BRNIPMON.exe 0x83bb9300 1 87 1 False 2009-12-11 19:21:28.000000 UTC N/A \D
evice\HarddiskVolume1\Program Files\Brownie\BRNIPMON.exe "C:\Program Files\Brownie\Brnipmon.exe" -BROTHER-PRINTER -
Brother HL-2170W series C:\Program Files\Brownie\Brnipmon.exe
(venv) root@f8123716479d:/forensics#
```

Exhibit A: Volatility 3 output for Terry's memory, highlighting PID 596 (Eraser.exe) and its associated -hide execution flag.

5.9 Identify a pattern of activity for Jo. Jo appears to have overwritten a large selection of files on one of the thumb drives. Are any of these files recoverable?

Forensic timeline analysis of the “work” USB storage device identified a suspicious mass-deletion event. At precisely 21:36:24 EST on November 20, 2009, more than 40 multimedia files were deleted within a single second—the speed of which is not consistent with manual user interaction. The original image data has been entirely overwritten by remnant data from PDF documents. Consequently, the original files are permanently destroyed and cannot be recovered.

This analysis was conducted under the authority of Search Warrant MT-2009-12-015-W001, which mandates the seizure and examination of all USB thumb drives in the subject's possession.

Artifact 1: The Timeline

- Timestamp: Fri Nov 20 2009 21:36:24 EST
- Affected Artifacts: A sequential cluster of 40+ multimedia files located in `/img_jo-work-usb-2009-12-11.E01/vol_vol2/`

Artifact 2: Data Overwrite Verification (Hexadecimal Analysis)

- Target File: /img_jo-work-usb-2009-12-11.E01/vol_vol2/p1/_SC00009.JPG
- MD5: 72423f07838807c1cb44a95240892fdf
- Expected Header: FF D8 FF E0 (Standard JPEG signature).
- Actual Hex: The raw data block contains PDF remnants and FF FF FF FF sequences which represent deleted or unused directory entries.

The screenshot displays the M57Kitty - Autopsy 4.22.1 interface. The top menu bar includes Case, View, Tools, Window, and Help. Below the menu is a toolbar with icons for adding data sources, viewing images/videos, communications, geolocation, timeline, discovery, generating reports, and closing cases. The left sidebar shows a tree view of the file system, with the following items highlighted:

- charlie-2009-12-11.E01_5426 Host
- charlie-work-usb-2009-12-11.E01_536921 Host
- jo-2009-12-11-001.E01_128369 Host
- jo-favorites-usb-2009-12-11.E01_431313 Host
- jo-work-usb-2009-12-11.E01_1 Host
- jo-work-usb-2009-12-11.E01_1 Host
- vol1 (Unallocated: 0-62)
- vol2 (Win95 FAT32 (0x0b): 63-255968)
- \$OrphanFiles (0)
- \$Unalloc (1)
- .fsevents (4)
- .fsevents (0)
- .Spotlight-V100 (3)
- .Trashes (4)
- HighQuality (0)
- Papers1 (96)
- Papers10 (96)
- Papers11 (96)
- Papers12 (96)
- Papers13 (96)
- Papers14 (96)
- Papers15 (96)
- Papers16 (96)
- Papers17 (96)
- Papers2 (96)
- Papers3 (96)
- Papers4 (96)
- Papers5 (96)
- Papers6 (96)
- Papers7 (96)
- Papers8 (96)
- Papers9 (96)
- XFER-11-20-2010 (8)
- vol3 (Unallocated: 255969-255999)
- terry-2009-12-11-001.E01_194681 Host
- terry-work-usb-2009-12-11.E01_526769 Host

The main pane shows a listing of files in the directory /img_jo-work-usb-2009-12-11.E01/vol_vol2. The table has columns for Name, S, C, O, Size, Flags(Dir), Created Time, MD5 Hash, and Modified Time. The file _SC00009.JPG is highlighted in red. The metadata for this file is shown in the bottom pane:

Metadata

- Name: /img_jo-work-usb-2009-12-11.E01/vol_vol2/_SC00009.JPG
- Type: File System
- MIME Type: application/octet-stream
- Size: 334501
- File Name Allocation: Unallocated
- Metadata Allocation: Unallocated
- Modified: 2009-11-11 09:16:56 EST
- Accessed: 2009-11-20 00:00:00 EST
- Created: 2009-11-20 21:36:24 EST
- Changed: 0000-00-00 00:00:00
- MD5: 72423f07838807c1cb44a95240892fdf

Exhibit A: Forensic timeline capture highlighting the identical 21:36:24 timestamp across all artifacts.

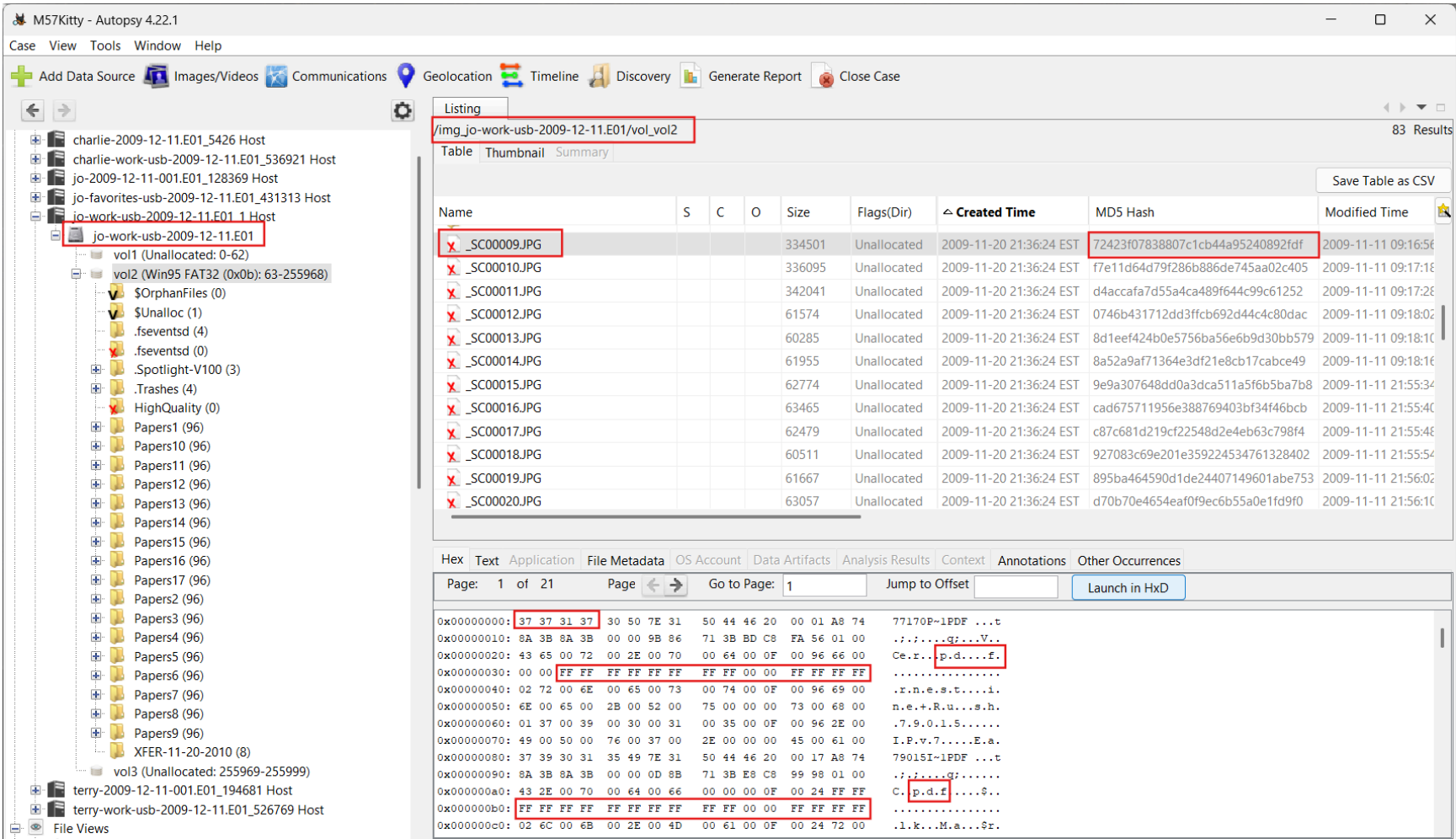


Exhibit B: Hex View capture of target file _SC00009.JPG, highlighting the text translation column demonstrating PDF-specific strings within a file system designated as a .JPG.

5.10 Identify a pattern of activity for Terry. Terry is involved in the sale of Jo’s old computer. How? Is there evidence of this on his machine?

Analysis of Terry Johnson’s workstation emails reveals a pattern of misappropriation and embezzlement due to a need to fund personal gambling. Using his administrative access, Terry decommissioned corporate hardware—specifically the workstation assigned to Jo Smith—under the false pretense of proper disposal. Email communications recovered from the workstation demonstrate that the user Terry actively deceived the company CEO regarding the disposal of this equipment. Instead of destroying or recycling the hardware, Terry sold Jo's intact computer for \$1,000 via a Craigslist advertisement linked to a personal email address (t93940@gmail.com). Digital contact records and email correspondence directly tie Terry to Aaron Greene, the buyer who then discovered the illicit material and initiated the police investigation. The extraction of this evidence is legally authorized under the voluntary consent provided by M57 Patents CEO Pat McGoo.

Artifact 1: Contact Record

- File Path: /img_terry-2009-12-11-001.E01/vol_vol2/Users/terry/Contacts/Aaron Greene.contact
- MD5 Hash: cbea75fe025eef1c74856b844595b7f9
- Content: Contains the buyer's name "Aaron Greene" and email address aarongreene12@gmail.com. Created 2009-11-30 12:59:34 EST.

Artifact 2: Deleted Craigslist Email

- File Path: /img_terry-2009-12-11-001.E01/vol_vol2/Users/terry/AppData/Local/Microsoft/Windows Mail/Local Folders/Deleted Items/1CFA4233-0000001E.eml
- MD5 Hash: 4c24e5f93d1971f8014949b60a892e63
- Content: Automated confirmation of the Craigslist posting titled "Dell Computer For Sale".

Artifact 3: Email Narrative

- Acquisition: Email from Terry to Jo arranging to swap her "broken down computer."
 - File Path: /img_terry-2009-12-11-001.E01/vol_vol2/Users/terry/AppData/Local/Microsoft/Windows Mail/Local Folders/Sent Items/3CCD36FA-00000005.eml
 - MD5 Hash: 76025ca017df9e4bbce4ece87f1b250c
- Deception: Email response to CEO Pat McGoo regarding equipment disposal: "Don't worry - I'll take care of everything."
 - File Path: /img_terry-2009-12-11-001.E01/vol_vol2/Users/terry/AppData/Local/Microsoft/Windows Mail/Local Folders/Sent Items/1F064827-00000006.eml
 - MD5 Hash: bbd1e1e032a3d0ecc91902f83c621139
- Hardware Verification: Email response to Pat McGoo confirming the serial number of the missing computer was "C1111".

- File Path: /img_terry-2009-12-11-001.E01/vol_vol2/Users/terry/AppData/Local/Microsoft/Windows Mail/Local Folders/Sent Items/162843B0-0000001A.eml
- MD5 Hash: d618348fc254597cb13bf1f5a28b5def
- Motive: Email indicating a need for "cash tonight" to enter \$50 buy-in poker tournaments.
- File Path: /img_terry-2009-12-11-001.E01/vol_vol2/Users/terry/AppData/Local/Microsoft/Windows Mail/Local Folders/Sent Items/76246241-00000018.eml
- MD5 Hash: 252078581468f6cf285a7f0ba91e8ebe

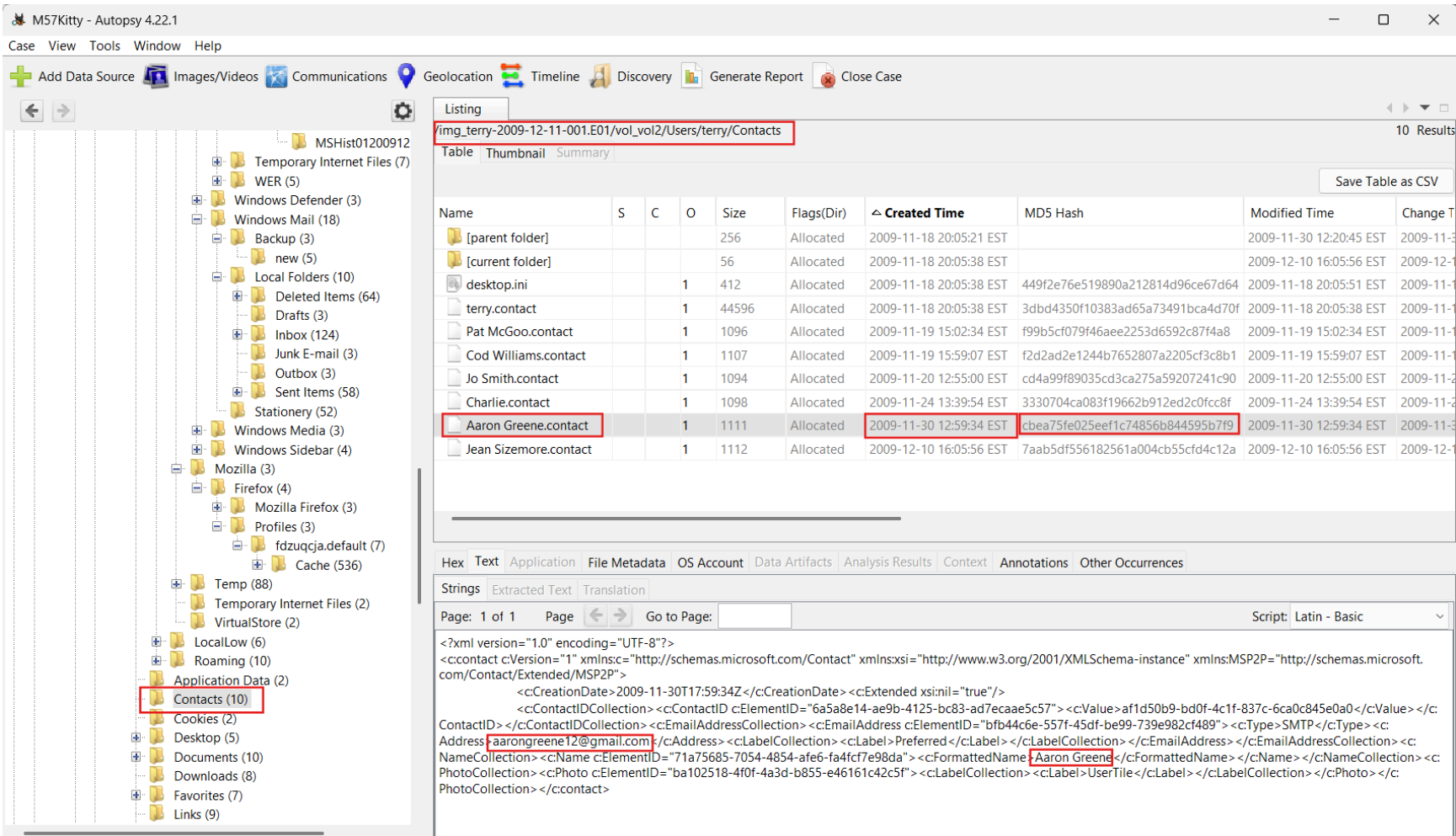


Exhibit A: Aaron Greene contact file.

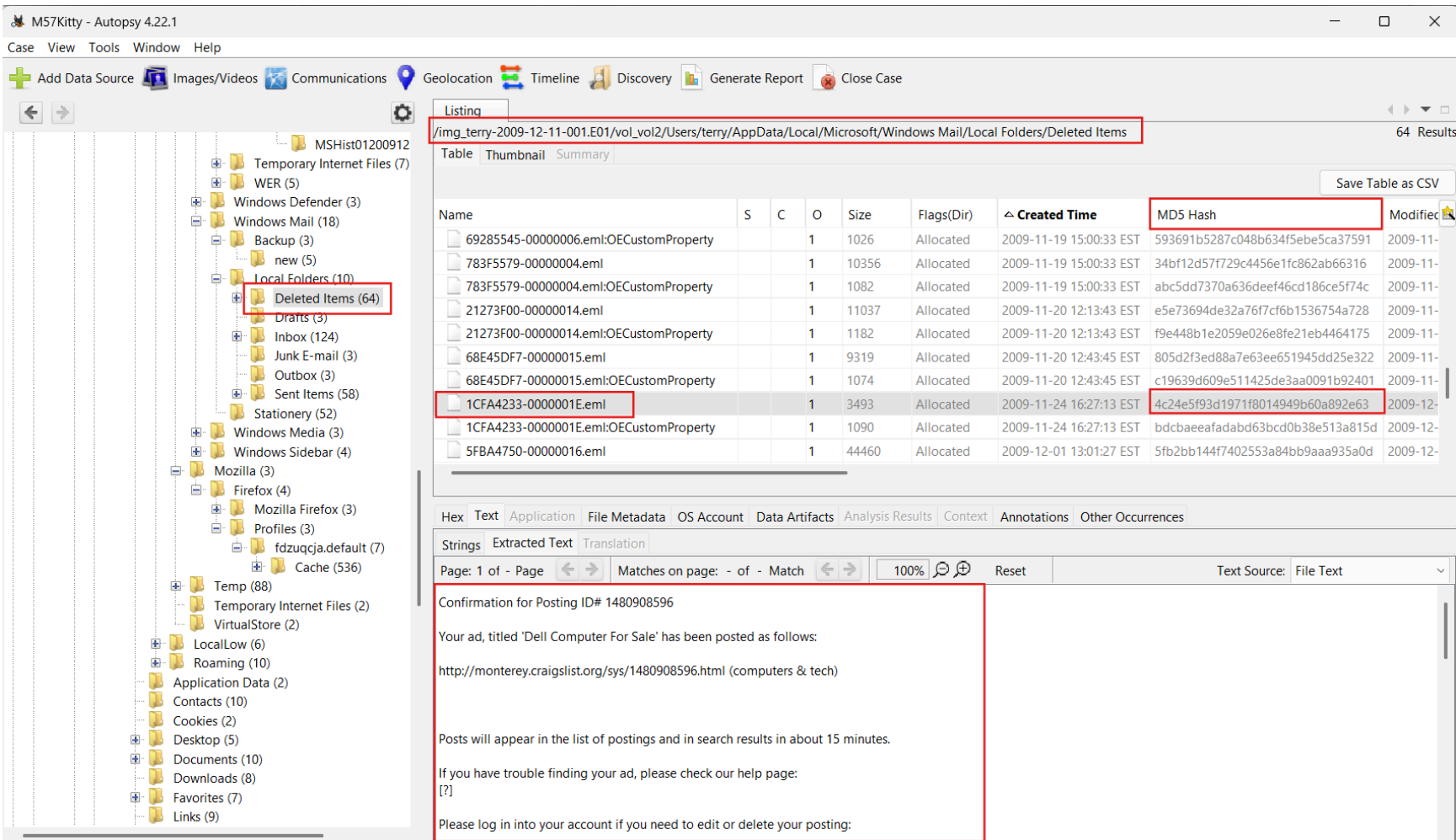


Exhibit B: Craigslist confirmation email from Terry's Deleted Items folder.

5.11 Identify use of evidence elimination tools. Which tools have been used, and by whom? Have they been successful?

Analysis across the M57 workstations and removable media reveals that there were attempts to eliminate evidence of their illicit activities. Two employees, Terry Johnson and Jo Smith, utilized distinct methods and tools with varying degrees of success. The documented use of evidence elimination tools—specifically Terry's hidden execution of Eraser.exe during the police raid and Jo's mass-overwriting of USB data—moves their conduct beyond mere possession or policy violation into the realm of active destruction of evidence.

1. Terry Johnson

Terry Johnson actively deployed commercial anti-forensic and data-wiping utilities. System registry analysis confirms he installed and executed CCleaner on November 24, 2009, and Eraser (a secure file-wiping tool) on December 10, 2009.

- Success Rate: Unsuccessful. Despite possessing these advanced tools, Terry's primary attempt to destroy the incriminating Craigslist emails was executed poorly. He merely soft-deleted the emails within Windows Mail, leaving them fully recoverable in the Deleted Items folder. While Eraser.exe was found actively running as a hidden process in RAM during the police seizure, it failed to destroy the most critical evidence linking him to the hardware embezzlement before the system was captured.

2. Jo Smith (The Overwrite Approach)

Jo Smith utilized a combination of file system deletion and physical data overwriting on removable media to eliminate evidence, alongside TrueCrypt to hide active data. On November 20, 2009, Jo executed an automated mass-deletion of over 40 images on the "work" USB drive.

- Success Rate: Mixed (Partially Successful). Jo was highly successful in destroying the files on the USB drive. Hexadecimal analysis of the deleted files (e.g., _SC00009.JPG) confirms the original image data was permanently overwritten with remnant data from PDF documents, rendering recovery impossible. However, Jo was unsuccessful in sanitizing the local workstation; a copy of the illicit video (KittyMontage.mov) was found fully intact and recoverable within the local drive's \$OrphanFiles directory.

Terry Johnson's Tools & Artifacts

- To view the evidentiary artifacts demonstrating Terry Johnson's failed attempt to delete the Craigslist communications via Windows Mail, see Exhibits A under section 5.5.
- For documentation of the 'Eraser.exe' wiping utility actively running in volatile memory during the police seizure, refer to the Volatility 3 process tree shown in Exhibit A under section 5.7.
- The system registry logs confirming the installation and execution of CCleaner and Eraser prior to the police raid are documented in the AppCompatCache analysis provided in Exhibit A under section 5.3.

Jo Smith's Tools & Artifacts

- The hexadecimal analysis proving Jo Smith successfully overwrote the deleted image files with PDF data is fully documented in Exhibits A and B under section 5.9.